RANJITH N

Tirupur, Tamil Nadu, India | 7604960261 | ranjith.n.work@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Science graduate (B.Sc, Data Analytics) building agentic, multi-agent AI systems with LangGraph, Retrieval-Augmented Generation (RAG), vector embeddings, and human-in-the-loop LLM orchestration. Skilled in prompt engineering, structured output validation, REST API design, and full-stack delivery.

EDUCATION

Bachelor of Science, Computer Science with Data Analytics: KPR College of Arts, Science and Research, 2023–2026

HSC — Computer Application & Accounting: Saradha Vidhyalaya Mat. Hr. Sec. School, 2022–2023

PROJECTS

AI Recruiting Pipeline Assistant *Personal Project*

- Built an agentic, multi-agent recruiting pipeline (Resume Parser, JD-Match, Bias Checker, Interview Question Generator, Ranking) on LangGraph's StateGraph with typed Pydantic schemas, RAG-grounded scoring (FAISS + sentence-transformers), and a hard human-in-the-loop approval gate.
- Shipped a FastAPI backend (Groq LLM, pdfplumber PDF ingestion) + React/TypeScript dashboard end-to-end.

Tech: Python, LangGraph, LangChain, FAISS, Sentence-Transformers, Groq API, FastAPI, Pydantic, React, TypeScript.

Agentic RAG — Multi-Step Reasoning Document Q&A System *Personal Project*

- Built an agentic RAG system where an LLM-driven agent decides when retrieval is needed, self-checks retrieved chunks for relevance, and reformulates the query on failure (up to 3 attempts) before answering — reducing hallucination by declining out-of-scope questions instead of guessing.
- Shipped a full-stack app: FastAPI backend (/chat, /upload, /health) with dynamic PDF ingestion (incremental FAISS updates), paired with a React + Vite UI showing a live, expandable reasoning trace per response.

Tech: Python, LangChain (tool-calling agents), Groq API (gpt-oss-120b), FAISS, Sentence-Transformers, FastAPI, React.

LLM-Powered RAG System — PDF Document Q&A *Personal Project*

- Built a RAG pipeline (PDF_RAG_FAISS_GROQ) for natural-language Q&A over PDFs using FAISS semantic retrieval + Groq LLM grounded generation; documented architecture and published on GitHub/LinkedIn.

Tech: Python, LangChain, FAISS, Vector Embeddings, Semantic Search, Groq API.

Autonomous AI Research Agent *Personal Project*

- Built a LangChain ReAct agent that reasons over a query, dynamically selects tools (Wikipedia API, DuckDuckGo), and auto-generates structured academic-style research reports end-to-end.

Tech: Python, LangChain (ReAct), Groq LLM, Agentic AI, Tool Use, Prompt Engineering.

Conversational AI Assistant — Gemini API *Personal Project*

- Built a context-aware conversational app using Gemini 1.5 Flash API with prompt engineering for consistent multi-turn responses.

Tech: Python, Gemini 1.5 Flash, LLM APIs, Prompt Engineering, Conversational AI.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, HTML/CSS, SQL

AI / LLM Engineering: LangChain, LangGraph, Agentic AI, Multi-Agent Systems, AI Agents (ReAct), RAG, Vector Databases (FAISS), Vector Embeddings, Semantic Search, Sentence-Transformers, Human-in-the-Loop (HITL) Design, LLM Orchestration, LLM APIs (Groq, Gemini), Prompt Engineering, Structured Output Validation, Model Evaluation

Frameworks & Tools: FastAPI, PostgreSQL, Git, GitHub, MySQL, Figma, Render.com

Core Concepts: REST API Development, Data Analytics

SOFT SKILLS

Communication, Team Collaboration, Leadership, Problem Solving, Adaptability, Quick Learner, Enthusiasm

AREA OF INTEREST

Game Development, Animation

CERTIFICATIONS

- Prompt Engineering for Developers — DeepLearning.AI
- Data Analytics Job Simulation — Deloitte (via Forage), April 2026

ACHIEVEMENTS

- Asian Book of Records
- District ThrowBall Winner (Captain)